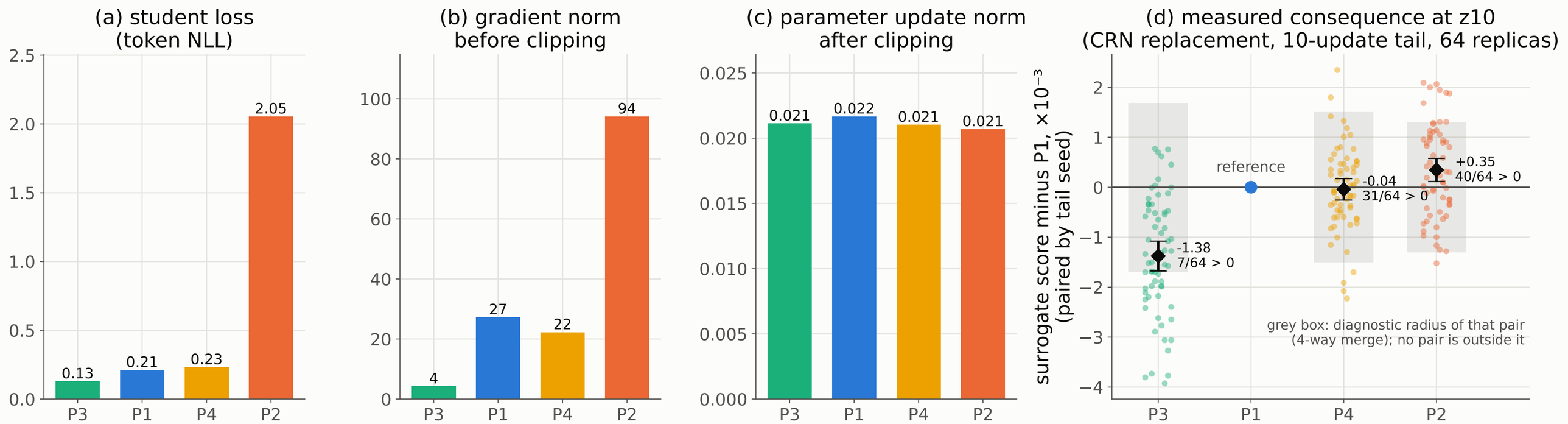
(a) student loss
(token NLL)
2.5
2.0
1.5
1.0
0.5
0.0
0.13
0.21
0.23
2.05
P3
P1
P4
P2
(b) gradient norm
before clipping
100
80
60
40
20
0
4
27
22
94
P3
P1
P4
P2
(c) parameter update norm
after clipping
0.025
0.020
0.015
0.010
0.005
0.000
0.021
0.022
0.021
0.021
P3
P1
P4
P2
(d) measured consequence at z10
(CRN replacement, 10-update tail, 64 replicas)
surrogate score minus P1, ×10⁻³
(paired by tail seed)
2
1
0
−1
−2
−3
−4
reference
-1.38
7/64 > 0
-0.04
31/64 > 0
+0.35
40/64 > 0
grey box: diagnostic radius of that pair
(4-way merge); no pair is outside it
P3
P1
P4
P2
P1 replay of K=20 demos
P2 teacher corrections on student states
P3 self-generated successes
P4 teacher rewrites of the demos